

PHP Learning Roadmap: Beginner to Advanced

1. Beginner Level

- **Introduction to PHP**
 - What is PHP? (Server-side scripting, PHP vs client-side languages)
 - Setting up a PHP environment (XAMPP/MAMP, Docker, PHP CLI)
- **Basic Syntax**
 - PHP tags (<?php ?>), comments, and output (echo, print)
 - Variables, data types (strings, integers, arrays, etc.), and type juggling
 - Constants (define(), const)
- **Operators & Control Structures**
 - Arithmetic, comparison, logical operators
 - Conditional statements (if, else, switch)
 - Loops (for, while, do-while, foreach)
- **Functions**
 - Built-in functions (string, math, array)
 - User-defined functions, parameters, and return values
 - Variable scope (global, static, local)
- **Forms & User Input**
 - Handling GET vs POST methods
 - Basic form validation and sanitization

2. Intermediate Level

- **Arrays & Strings**
 - Indexed, associative, and multidimensional arrays
 - Array functions (sort, array_map, array_filter)
 - String manipulation and regex basics (preg_match, str_replace)
- **File Handling**
 - Reading/writing files (fopen, fwrite, file_get_contents)
 - File uploads and security considerations
- **Error Handling**
 - Error types (notices, warnings, fatal errors)
 - Custom error handling (set_error_handler, exceptions with try/catch)
- **Object-Oriented Programming (OOP)**
 - Classes, objects, properties, and methods

- Inheritance, encapsulation, polymorphism
- Magic methods (`__construct`, `__destruct`)
- **Namespaces & Autoloading**
 - Organizing code with namespaces
 - Composer basics and autoloading (PSR-4)
- **Databases & MySQL**
 - Connecting to databases (PDO, MySQLi)
 - CRUD operations, prepared statements (security against SQL injection)
 - Intro to ORMs (e.g., Eloquent)
- **MVC Architecture**
 - Separation of concerns (Model, View, Controller)
 - Building a simple MVC app from scratch
- **SOLID Principles**

Naming convention

3. Advanced Level

- **Security Best Practices**
 - Preventing SQL injection, XSS, CSRF
 - Password hashing (`password_hash`, `password_verify`)
 - Input validation/sanitization (`filter_var`, HTML escaping)
- **Sessions & Authentication**
 - Session management (`session_start`, cookies)
 - Building login systems, role-based access
 - OAuth integration (e.g., Google/Facebook login)
- **Working with APIs**
 - Consuming REST APIs (`cURL`, `file_get_contents`, Guzzle)
 - Building RESTful APIs (JSON responses, JWT authentication)
- **Advanced OOP Concepts**
 - Traits, interfaces, abstract classes
 - Type hinting, anonymous classes, and closures
- **Design Patterns**
 - Singleton, Factory, Observer, Dependency Injection
 - MVC in-depth
- **Performance Optimization**
 - Opcode caching (OPcache)

4. Miscellaneous

- HTTP Verbs
- HTTP Response

